RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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DEPARTMENT OF MATHEMATICS

Academic year 2024-2025 (Odd Semester)

Date	**02-01-2025**	**Time**	**10:00 AM-11:30 AM**	
Test	**Improvement Test**	**Maximum Marks**	**50**	
Course Title	**Fundamentals of Linear Algebra, Calculus & Differential Equations**		**Course Code**	**MA211TB**
Semester	**I**	**Programs**	**AS, CH, IM and ME**	

Sl. No.	**Answer all questions**	M	CO	BTL
1(a)	Solve the problem of undamped forced vibration of a spring governed by the differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = x^2e^{3x}$.	6	4	3
1(b)	If $z = 3x^2 - 4y^3 - 7x^2y^3$, then prove that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$.	4	1	1
2	Solve the differential equation $\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 4x = 3\sin(t) + 4\cos(t)$, when $x(0) = 1$ and $x'(0) = 0$.	10	3	3
3	Using variation of parameter method, solve $\frac{d^2y}{dx^2} + 4y = \tan(2x)$.	10	3	3
4	Solve $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 5y = x^2\sin(\ln(x))$.	10	2	2
5(a)	Prove that $V = e^{3x+4y}\cos(5z)$ satisfy the Laplace's equation $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$.	5	2	2
5(b)	If $u = f(r,s,t)$ where, $r = \frac{x}{y}, s = \frac{y}{z}$ and $t = \frac{z}{x}$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$.	5	1	2

BTL-Blooms Taxonomy Level, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	9	15	20	6	4	20	26	-	-	-

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